

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tyger Valley

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Group: 4

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Indicate	Yes	No
Plagiarism report attached		

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Signature 	Date 21 June 2024
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Section A

Question 1

Windows share permissions are only applied to folders retrieved over a network, resulting in no protection against users accessing the folder locally from the device hosting it over the network. These permissions are inherited by all files and subdirectories as these permissions are set in the root of the share. Not only can permissions be used with NTFS, but it can also be used with FAT and FAT32 file systems. This is can only be configured through “Advanced Sharing” properties in the permission settings (Murphy, 2024).

These are the following Windows share permission types:

- Full control
- Change
- Read (Murphy, 2024)

NTFS permissions are applied to files and folders locally and on a network. NTFS permissions are often assigned to security groups for multiple users; however, it is possible to apply directly to user accounts. All files and subfolders of a folder inherit all NTFS permissions assigned to that folder. This can be configured through the security tab from the properties of the file or folder.

These are the following NTFS permission types:

- Full control
- Modify
- Read and execute
- Read
- Write
- List folder contents (Murphy, 2024)

Due how configurable the NTFS permissions are, it can be used in conjunction with share permissions to provide better flexibility for permissions. Furthermore, if both NTFS and share permissions are applied to the same file or folder, the most restrictive permission will be applied when it is accessed over a network. However, if the file or folder can be accessed locally then only the NTFS permissions are applied (Köller, 2023).

Question 2

The mainstream types of operating systems include Network Operating System, mobile and client operating system.

A Network Operating System (NOS) is an operating system that is used to run servers especially for business networks. The features associated with NOS include allowing users to share resources with each other and improved functionality of users, groups, programs as well as management data (Daisy, 2023).

The benefits of NOS consist of hardware and software upgrades that can be easily implemented, and how stable the server is. Additionally, the server can be utilised from several locations with various systems.

The detriments of NOS is seen through the high cost to purchasing and running the server and that the server requires recurring updates and maintenance (Daisy, 2023).

Transforming a client-server out of NOS allows it to become an invaluable tool for educational environments. The server with NOS can act as a centre for all computers within an educational facility where students and educators can easily share learning resources amongst each other. This will lead to an increase in productivity and allow all students to have access to the necessary learning material if they don't possess the necessary equipment. The drawback to this is that students are not able to access the server and its resources away from the educational environment and is a costly venture for the facility as it will need to maintain both the server and computers used by the students.

A mobile operating system is an operating system that was designed to work over mobile devices with a touch-operated interface like cell phones and tablets. This operating system provides a seamless and user-friendly interface for users due to the touchscreen optimisation (I.T.VEDANT, 2024).

The benefits of a mobile operating system include how highly customisable it is, which allows users to configure their device to their own preferences. Moreover, it integrates with cloud services allowing users to easily backup data and sync information between multiple devices.

The detriment of a mobile operating system is shown as the hardware that the operating system is associated with is limited compared to computers such as less overall processing power and performance (I.T.VEDANT, 2024).

Compared to the previous operating system types, the mobile operating system is the most accessible type of operating system as the mobile hardware that it is installed on is owned by 80% of the world's population (Markovic, 2024). The accessibility of this operating system can prove to be beneficial for educational environments as it allows students to easily access important learning resources and information from educators. Additionally, the syncing abilities that the operating system can provide introduces an advantage to learning, which is allowing students to access academic information from another device while away from their learning institution. This presents another level of accessibility to academic resources for students, improving the overall learning process. However, due to the hardware limitations, mobile operating systems are not suitable for

completing academic activities like projects when compared the hardware related to the other types of operating systems.

A client operating system can be defined as two distinct versions, which is a business client operating system that is used to work as a client in a managed business domain network and a home client operating system that is used to work as a client in a network in a small area like a home environment or as a standalone client. This type of operating system is usually found on desktops and laptops and occasionally mobile devices.

The benefits of a client operating system include how it is designed to be compatible with various types of hardware components such as printers, keyboards, mice and cameras (Infoventure, 2023). Furthermore, the interface that is provided allows users to complete complex tasks like creating documents easier than other operating system types.

Like the mobile operating system, the client operating system can also be used to receive learning resources from educators. However, the hardware accompanied with the client operating system is not restricted by hardware limitations and in conjunction with its user-friendly interface, students can complete academic activities and tasks more efficiently. This increase in efficiency allows students to become more productive in a learning environment and consume more content easier. However, most devices that house client operating systems are not portable resulting in the learning information not being as easily accessible as devices with mobile operating systems.

If the learning institution has the necessary resources to provide access to a server for all students, then the NOS should be optimised for a learning environment. This is due to the ability for students and educators to easily share academic resources with each other and provides the possibility for students to collaborate with each other to complete tasks. Furthermore, providing the resources also allows less fortunate students to be on the same level as students who have their own resources for education away from the academic facility. However, if the learning institution cannot provide those resources, then the devices with the mobile operating systems should be optimised for learning. Optimising these devices would be beneficial for students due to the device's portability and overall accessibility.

Question 3

3.1) Creating complex passwords for both ordinary and administrative accounts with methods such as increasing the length of the password and incorporating various symbols and numbers leads to passwords that are harder to crack. Since all passwords can be cracked, the goal of complex passwords is to make cyber criminals abandon their attempt to hack these accounts due to the cracking process taking too long (ENZOIC, 2024). Complex passwords can also protect against unauthorised users within an organisation attempting to gain access to administrative accounts.

3.2) BIOS and UEFI passwords are a system user password that is needed before booting can occur in an operating system and can be used improve security of an individual computer. The setup program for BIOS and UEFI can be used to configure the system password. This password is effective at preventing unauthorised users from tampering with the settings for hardware and software on a computer. However, if the someone has physical access to the computer case, they can use it to reset the password (Hoffman, 2014). Organisations can make use of these passwords to limit the changes that employees can make on the devices, preventing any potential damage from happening.

3.3) There are many malware vectors and the most common include:

- Worms, which duplicates between multiple processes in the system memory.
- Viruses, which are hidden in the code of an executable process image and when the program file is executed the virus will also be executed with the same permissions.
- Boot sector viruses, which infects the code in a boot sector and tries to place itself into the computer memory.
- Trojans, which is malware that is hidden within a software that will seem to be legitimate to users and will be downloaded with the software and execute with the same permissions.

The CompTIA seven-step best practice procedure for malware remover consists of:

1. Investigate and verify malware symptoms.
 - Defining the type of malware that has infected the system determines the severity of the consequences as well as the steps that need to be taken in order to remove it effectively.
2. Quarantine infected systems.
 - Quarantining the device prevents the malware from communicating with the network that this is connected to. If this step is not done, then malware like worms can spread over the network and infect other devices.
3. Disable System Restore in Windows.
 - Disabling these backups from occurring prevents the malware from being backed up and infecting all other backups. It is safest to delete all backups that were made, however if it is imperative that they need to be recollected then an antivirus can be used to determine if it was infected.
4. Remediate infected systems:
 - a) Update anti-malware software.
 - b) Scanning and removal techniques.
 - a. Updating ensures that the anti-malware software is well-equipped to identify and perform removal techniques to clean the system from malware.

5. Schedule scans and run updates.
 - Scheduling consistent security scans confirms whether any new malware has infected the system so that it can be removed before it does too much damage. Updating the security of the operating system ensures that it will no longer be susceptible to older versions of malware.
6. Enable System Restore and create a restore point in Windows.
 - Enabling System Restore and creating a restore point is vital as a clean backup copy of the critical operating system files is created and can be used to restore the system before an event like malware has occurred (Wright, 2024).
7. Educate the end user.
 - Educating the user of the consequences of malware and how to identify it equips the user with the knowledge to avoid infecting their devices in the future.

Question 4

4.1) Confidentiality refers to the information and data that only certain individuals can access. Organisations ensure that unauthorised individuals are unable to view and share this information. To ensure that confidentiality can occur over a SOHO network, methods of encryption and authentication is implemented to protect all the data stored or shared over the network. Scenarios that demonstrate the need for encryption to protect data in the SOHO network consist of medical reports in healthcare facilities and personal banking information.

Integrity refers to data remaining accurate as well as trustworthy when stored and transferred and that any modification can only occur when authorised. Integrity can be accomplished through integrating digital signatures and hashing in the SOHO network to determine if information was tampered with. This is essential to networks, because altered data can lead to negative consequences such as bad decisions and potential safety hazards. Integrity is vital for situations like users and banks receiving financial transactions and statements (Make Computer Science Great Again, 2023).

Availability refers to data and information remaining accessible for all users who are authorised to view and modify. Redundancy and failover mechanisms are implemented to ensure availability within a SOHO network. Availability is crucial for SOHO networks, because inaccessible data can lead to operational disruptions as well as financial damages (Make Computer Science Great Again, 2023). Availability is especially required for hospitals when individuals need to retrieve important medical information over the network or when individuals wish to see their banking information like their balance.

4.2) Wired Equivalent Privacy (WEP) is the original security standard that is meant to deliver a level of security to wireless networks that is similar to wired networks. This means that WEP supports older hardware that cannot support recent security protocols. WEP uses a method called RC4 symmetric cipher to encrypt and decrypt traffic on the network with a single 64- or 128-bit key. However, WEP has a massive cryptographic vulnerability as the key can be reused, which makes networks more susceptible to attacks. Furthermore, WEP is not a modern security standard, and it is recommended to upgrade to a more recent security protocol (Hanko, 2022).

Wi-Fi Protected Access (WPA) is a security protocol that was developed to fix the vulnerabilities introduced from the WEP security protocol. The WPA security protocol also uses RC4 symmetric cipher for encryption, but it uses Temporal Key Integrity Protocol (TKIP) to prevent the attacks that were established against WPA. Moreover, WPA introduced new security features such as allowing users to protect their networks with passwords. However, WPA is vulnerable to pre-shared key attacks as well as any attacks that start within the network (Hanko, 2022).

Unlike WPA, WPA2 uses the Advanced Encryption Standard (AES) cipher and Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP) instead of RC4 and TKIP. Replay attacks are harder to use on networks with the WPA2 security protocol, because CCMP delivers authenticated encryption. Additionally, it also uses pre-shared key (PSK) authentication that generates a key that encrypts networks with the use of a passphrase and can be applied for a group of users. Unfortunately, PSK authentication is vulnerable to attacks that tries to recover the passphrase so the passphrase must be fourteen characters or more to prevent these attacks.

When comparing WEP, WPA and WPA2 for SOHO networks, WPA2 is the most suitable security protocol, because it is the most secure and strongest among the other two protocols and can be used to provide security to groups of users within the company.

Question 5

5.1) Security policies for organisations are extremely important as it is essential to protecting sensitive information and data from malicious individuals. These policies can also ensure that the business meet the necessary requirements of any applicable regulations (ThriveDX Media, 2022). Security procedures are important as it provides an efficient way to implement and manage these security policies on the mobile devices. Furthermore, security training ensures that employees of the company fully understand all the security policies and procedures. This is important as employees can guarantee that they stay within regulations when working and are aware of the consequences of security vulnerabilities.

5.2) The authentication mechanisms for mobile devices occur on the lock screen and can consist of:

- Patterns:
 - This refers to the user creating a pattern with lines connecting to dots through swiping and the mobile device determines if the pattern matches the original pattern created by the owner of the device. This is one of the weaker forms of authentication for mobile devices.
- Passwords or a Personal Identification Number(PIN) authentication:
 - Passwords and PINs can be configured for the lock screen and is used to create encryption keys.
 - A PIN is most suitable for general users who wish to have satisfactory security for their personal device, however a strong password is recommended to users if their mobile device contains sensitive data and has a high compromise risk.
- Biometric authentication:
 - This type of authentication refers to any biometric features from the user of the mobile device.
 - Facial recognition uses the camera of the mobile device to create a template from a 3-D image of the user's face for authentication.
 - Fingerprinting is when the user scans their fingerprint, which is then used to confirm that it is the same as the template that was created during an enrolment fingerprint scan.

5.3) Bring your own device (BYOD) is when the mobile device is owned by the employee and can be used for work tasks if they agree to some oversight and auditing. It is very popular among employees, which improves employee happiness, and it is more cost effective for organisations. However, it leads to security vulnerabilities where data breaches can occur as the company are unable to fully manage the device (Applivery, 2023).

Corporate owned, business only (COBO) is when the device is owned by the organisation and can only be used for work. The benefits and downsides of this model are contradictory to BYOD as it leads to an increase in device and data security as well as more efficient device management, but it limits employee happiness and leads to higher costs for the company.

Corporate owned, personally enabled (COPE) is when the device is owned by the organisation and can be used by employees to perform personal activities. This provides better security and management than BYOD and increases employee satisfaction, but like COBO it leads to higher costs for the company.

Choose your own device (CYOD) is when employees can choose from a list of devices that their company offers and is similar to COPE. This provides substantial flexibility for employees and like COPE it has good security and management capabilities. However, like COBO and COPE it can be costly to employers as they own the devices.

5.4) Mobile Device Management (MDM) is a class of enterprise software that applies security policies in a business network to mobile devices. Mobile devices with this software can be configured by the business to restrict or allow specific applications and any built-in features such as the device's microphone and camera. Restrictions like these are used to protect important data and information found at sites from malicious employees. This software also ensures that the security on the mobile device, like the patch status and antivirus, are up to date and that the firewall is running and has been configured properly. MDM can locate any lost or stolen mobile devices to certify the device's retrieval, however if a security breach occurs then the device can be remotely locked or wiped. Another advantage of tracking the location of the device is that it can be configured to automatically restrict data when outside of an authorised area (Trio Team, 2024).

5.5) Remote backup applications like cloud services can be used to routinely create remote backups of data and applications from devices. The organisation can use these backups to retrieve and re-establish all the data on the mobile device that was lost or stolen. However, it does not protect the information located on the device against individuals who try to gain access to the device.

Locator applications use Global Positioning System (GPS) to track the location of devices that were stolen or lost, and it comes with features such as locking the device to prevent unauthorised access, disabling critical features like the wallet as well as ensuring network services stay enabled. However, if the device cannot be recovered then the data on the device can be remotely wiped. Examples of remote wiping includes device wiping, which performs a factory default reset on the device and enterprise wiping, which only removes corporate data.

5.6) The first course of action that should be used when troubleshooting applications that fails to launch, close, or crash is to use force stop on the application or disable it and attempt to re-open. Force stop and disable can be found in the Apps menu in the Settings application. If this does not fix the issue, then the app cache must be cleared in the same menu where the force stop and disable option can be found. If the issue persists then the device should be restarted, the application should be updated and if it cannot be updated then determine if it is still compatible with the current version of the device's operating system and that there is enough storage space. If after all the above steps and problem still exists, then the application must be deleted and re-installed.

5.7) One of the first approaches towards solving WI-FI connectivity issues is to position the mobile device closer to the access point and holding it in a different way to try and mitigate any potential interferences. I can also determine if the device is on airplane mode and if it is then it must be switched off. If the issues persist then I would enter the settings application to access the networks menu and forget the specific network that the mobile device is attempting to connect to and re-connect to it. I can also determine if the access point supports the same 802.11 standard as the mobile device and if it doesn't then the access point must be configured into compatibility mode. If

I am able to, I would perform an operating system or firmware update for both the mobile device and access point to conclude if it fixes the problem.

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